

Safety

The rules that do not bend

Read this completely before Lesson 10. Post it on the wall.

WHAT YOU ARE DEALING WITH

A neon-sign transformer turns household power into 9–15 kV at 30–60 mA. Roughly 10 mA across the chest can stop a heart, and 100 mA very reliably does. This supply delivers several times that, it is a *current-limited but not current-safe* source, and it will not let go of you — above about 15 mA your hand muscles clamp shut on the conductor. The tank capacitor stores a lethal charge **after the power is off** and will hold it for a long time. Nothing in Lessons 1–9 can hurt you. Everything from Lesson 10 can kill an adult.

The five rules

- 1 The adult owns the plug.** One adult, named in advance, is the only person who plugs in, switches on, or decides that the machine may be energized. That does not rotate, and it does not get delegated because someone is busy. Everyone else treats the machine as live until that person says otherwise.
- 2 Unplug, then short, then touch.** Never in another order, and never skipping a step. Pull the plug from the wall — not the switch, the plug. Then take the grounding stick and short the capacitor terminals together, and *leave it shorted* while your hands are inside. A capacitor that was discharged five minutes ago can recover a dangerous voltage on its own through dielectric absorption. Short it again every time.
- 3 One hand.** When anything is or might be energized, your other hand goes in your pocket or behind your back. Current that enters one hand and leaves the other crosses your heart. Current that enters and leaves the same hand mostly does not. This is a habit built in Lesson 3, when it costs nothing, precisely so it is automatic in Lesson 11.
- 4 Never alone.** The machine is never energized, or capable of being energized, with one person in the room. The second person's job is to be able to pull the plug and call for help. They should know where the plug is without looking.
- 5 Stand back and stay back.** Once it is running, nobody is within arm's reach of any part of it. Adjustments happen with the plug out and the capacitor shorted. There is no adjustment worth making while it runs.

Build these in, not on

Grounding stick	A length of dry hardwood or PVC with a metal hook at the end, a heavy wire from the hook to your RF ground. Build it <i>first</i> , before any high-voltage part is assembled. It is the single most important object in the room.
Bleeder resistors	Every capacitor in the bank gets a high-value resistor permanently across it, so the bank drains on its own in seconds rather than holding a charge for hours. Bleeders are a backup for the grounding stick, never a replacement for it.
Safety gap	A fixed spark gap wired across the transformer output, set slightly wider than the main gap. The coil's tank circuit can reflect voltage spikes back into the transformer and destroy it; the safety gap fires first and shunts them. It protects the equipment, and equipment that fails unexpectedly is a hazard to people.
Interlocked enclosure	The transformer, capacitor and gap live inside a box. The box has a switch that cuts mains power when it is opened. Test the interlock deliberately before you trust it.
A real switch	A proper switched and fused enclosure with an indicator lamp, at a distance from the coil. Not a power strip on the floor.
RF ground	The secondary's base wire goes to <i>its own ground rod</i> driven outside, not to the mains earth in a wall outlet. Streamers returning through house wiring will destroy electronics elsewhere in the building and can energize things that should not be energized.

The other hazards, which are easy to forget

Pacemakers and implants	A running Tesla coil produces intense electromagnetic fields. Anyone with a pacemaker, implanted defibrillator, insulin pump, cochlear implant or hearing aid must not be in or near the room. This is not a precaution to weigh against convenience — ask everyone before you fire it, every time.
Electronics	Phones, watches, laptops, hearing aids and car keys go in another room. Not in a pocket. The coil does not distinguish between an antenna and a phone.
Radio interference	A spark-gap coil is a broadband radio transmitter, and an inefficient one. It will interfere with radio, television and wireless service well beyond your own house, and unintentional radiators are regulated. Run it in short bursts, keep total runtime low, and stop if a neighbour reports interference.
Ozone and nitrogen oxides	Sparks in air make both, and both are lung irritants. Ventilate the space, and do not run it in a closed basement room. If you can smell it strongly, stop and air the room out.
Ultraviolet	An arc emits UV. Do not stare into the gap. Safety glasses, and the gap enclosed or shielded.
Noise	A spark gap is genuinely loud and the sound is impulsive. Hearing protection for everyone in the room.
Fire	You are making sparks near wood, dust and insulation. A fire extinguisher within reach, a clear bench, and nothing flammable near the gap.
Burns	Streamers are hot enough to burn skin instantly and to ignite paper. The RF burn from a streamer is deep and does not hurt at first, because it destroys nerve endings on the way in.

Before every run — read it aloud

1. Is the grounding stick in the room and within reach?
2. Is the capacitor bank shorted right now, and are my hands out?
3. Is the enclosure closed and the interlock tested?
4. Is the RF ground connected to the ground rod, and the connection tight?
5. Has everyone with an implant left, and are all electronics out of the room?
6. Is there a second person, and do they know where the plug is?
7. Safety glasses and hearing protection on everyone?
8. Is the extinguisher within reach and the bench clear?
9. Is everyone at least an arm's length beyond the longest streamer I expect?
10. Am I the named adult, and am I the one about to close this switch?

IF SOMEONE IS SHOCKED

Do not touch them. **Kill the power first** — pull the plug — because if they are still in contact with the conductor you will be the second casualty. Then call emergency services. Any electrical shock across the chest warrants medical assessment even if the person feels fine afterwards: electrical injury causes heart rhythm disturbances that can appear hours later, and internal burns that are invisible from outside. “He said he was fine” is not a clinical assessment.

Why the rules are written this way

Every rule above exists because it is a habit, not a judgement. In the moment that something goes wrong you will not reason your way to the right action — you will do whatever you have always done. That is the entire reason Lessons 1–9 use batteries that cannot hurt anyone and still insist on one hand, discharge before touching, and a second person in the room. The rules are the point of the low-voltage lessons; the physics is a bonus.

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